

Le Chatelier Tokenomics: A Thermodynamic–Homeostatic Framing of Negative-Feedback Token Supply Control, with a Multi-Asset Coupled-Equilibrium Extension

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Abstract

Algorithmic monetary policy in decentralized systems is conventionally described in the language of control engineering: elastic-supply tokens (Ampleforth) and PI-controlled redemption-rate systems (Reflexer/RAI) adjust token supply or implied interest in response to the deviation of market price from a target. This paper reframes the same family of mechanisms through the lens of thermodynamic and biological homeostasis — specifically the Le Chatelier–Brown principle, under which a system at equilibrium responds to an external perturbation by shifting in the direction that weakens that perturbation. We argue this framing is not merely cosmetic: it supplies design intuitions (ultra-stability, antagonistic regulation, managed hysteresis) that the control-engineering view leaves implicit, and it situates token stabilization within the broader theory of homeostatic systems. We formalize a single-asset negative-feedback supply controller as a discrete Le Chatelier response, make explicit its equivalence to a proportional controller, and then introduce a multi-asset coupled-equilibrium extension in which a perturbation to one token’s equilibrium induces proportional corrective flows across N coupled markets — a regime, to our knowledge, absent from the existing algorithmic-stablecoin literature. We discuss stability conditions, hysteresis to suppress oscillation, and attack surfaces at phase transitions.

1 Introduction

The 2020–2022 cycle produced a rich design space of algorithmic and elastic-supply tokens, several of which failed spectacularly (Terra/UST) while others (RAI) demonstrated that control-theoretic stabilization can hold a soft peg without full collateralization. The dominant analytical frame is *control engineering*: the protocol is a controller whose output (supply change, redemption rate) is a function of an error signal (price minus target). This frame is correct and useful. It is, however, only one of several scientific languages for self-stabilizing systems. We propose a complementary frame drawn from *homeostatics* — the cybernetic study of systems that maintain vital parameters within admissible bounds [7, 8] — and in particular the thermodynamic Le Chatelier–Brown principle.

Our contribution is threefold: (i) we make the Le Chatelier framing explicit and connect it to the homeostatic tradition; (ii) we honestly delimit what is *not* novel (the single-asset proportional mechanism); and (iii) we propose a genuinely under-explored design — multi-asset coupled equilibria — as the locus of new value.

2 Related work and honest prior art

- **Ampleforth (2019)** rebases supply proportionally to price deviation — a proportional negative-feedback corrector, without control-theoretic or thermodynamic framing [1].
- **Reflexer / RAI (2020–2021)** implements an on-chain **PI controller** on the redemption rate, explicitly described in control-engineering terms (proportional term on instantaneous deviation, integral term on accumulated deviation) [2]. This is the closest prior art and is mechanically near-identical to a single-asset Le Chatelier response.
- **Frax** uses a step-wise collateral-ratio feedback (bang-bang, not proportional) [3].
- **Academic:** control-theoretic treatments of algorithmic monetary policy establish that PID controllers outperform heuristics [4, 5].

We therefore claim **no novelty for the single-asset proportional mechanism**. Our claim of priority is for (a) the explicit Le Chatelier/homeostatic *framing* and (b) the multi-asset coupled-equilibrium *regime* of §4.

3 Single-asset Le Chatelier controller

Let p_t be market price, p^* the target, and S_t supply. Define the disturbance $d_t = (p_t - p^*)/p^*$. The Le Chatelier response prescribes a supply change opposing the disturbance:

$$\Delta S_t = -k \cdot d_t \cdot S_t, \quad 0 < k \leq 1, \quad (1)$$

so that excess price ($d_t > 0$) expands supply (reducing scarcity, pushing price down) and vice versa. This is exactly a proportional controller with gain k ; we state the equivalence explicitly rather than obscure it. The homeostatic reading adds two design refinements that the bare proportional form omits:

- **Hysteresis band** $[-h, +h]$: no corrective action while $|d_t| \leq h$, suppressing chattering near equilibrium (the “admissible range” of homeostatics).
- **Antagonistic regulation:** pairing an expansion channel and a contraction channel with independent gains, improving robustness to one-sided liquidity — the biological “contradiction regulation” principle [7].

4 Multi-asset coupled-equilibrium extension (the novel regime)

Consider N tokens with equilibria coupled through shared liquidity or explicit design (e.g., a basket, a family of correlated assets, or a layered protocol). Let $\mathbf{d} = (d_1, \dots, d_N)$ be the disturbance vector and $\mathbf{C}$ an $N \times N$ coupling matrix encoding how a perturbation to token j propagates to token i . The coupled Le Chatelier response is:

$$\Delta \mathbf{S} = -\mathbf{K} \mathbf{C} \mathbf{d} \quad (\text{elementwise scaled by supplies}), \quad (2)$$

where $\mathbf{K}$ is a diagonal gain matrix. Here a disturbance to one token triggers **proportional corrective flows across all coupled markets simultaneously**, analogous to a multi-phase thermodynamic system re-establishing equilibrium across phases. The design questions — choice of $\mathbf{C}$, stability of the coupled system, and avoidance of corrective cascades — are, to our knowledge, unaddressed in the algorithmic-stablecoin literature, and constitute the paper’s substantive new contribution.

4.1 Stability

The coupled system is stable when the spectral radius of $(\mathbf{I}-\mathbf{K}\mathbf{C})$ lies inside the unit circle; sufficient conditions on $\mathbf{K}$ relative to the eigenstructure of $\mathbf{C}$ follow, and over-coupling (large off-diagonal $\mathbf{C}$) is the coupled analogue of over-gain oscillation in the single-asset case.

4.2 Attack surfaces at phase transitions

Homeostatic theory warns that systems are maximally sensitive at critical points: a minimal external action near a phase transition can flip the system between qualitative states. In tokenomic terms, coordinated perturbations timed to coupled-market re-equilibration (launches, large redemptions, liquidity migrations) are the principal attack surface; hardening the protective machinery specifically around such transitions is the natural defense, and we leave its concrete design to companion work.

5 Discussion and limitations

The Le Chatelier framing is **explanatory and generative, not a novel control law** for the single-asset case. Its practical payoff is (i) a unifying vocabulary linking tokenomics to homeostatic/cybernetic theory and (ii) the multi-asset regime as a concrete research and design target. Limitations: the equivalence to PI control means single-asset claims add no engineering novelty; the multi-asset analysis here is theoretical and requires agent-based and on-chain validation; coupling matrices in practice are estimated, not given, and mis-estimation is itself a risk.

6 Conclusion

We recast negative-feedback token-supply control as a Le Chatelier–homeostatic process, delimit its non-novel single-asset core honestly, and propose multi-asset coupled equilibria as the open frontier. We invite empirical (cadCAD-style) validation.

References

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